Supplementary Information

Significantly Accelerated Hydroxyl Radicals Generation by Fe(III)-Oxalate Photochemistry in Aerosol Droplets

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Supplemental Experimental Details

Section S1. Materials and reagents

$FeCl_3$, $Na_2C_2O_4$, HCl, NaOH, Na_2SO_4, acetone, isopropanol, acetonitrile and phosphoric acid were all of analytic grade and purchased from Sinopharm Chemical Reagent Co., Ltd. Benzoic acid (BA, 99%) and *p*-Hydroxybenzoic acid (*p*-HBA, 99%) were supplied by Aladdin China Company. Hydrophobic (alkyl treated) fumed silica (CAB-O-SIL TS720) was purchased from Cabot Corp., Boston, MA. N_2 and air with ultra-high purity of 99.999% and SO_2 (2.21×10^{15} molecule cm^{-3}, balanced by N_2) were obtained from Shanghai TOMOE gases Co., LTD. The super-hydrophobic quartz wafers (water contact angle, 144° ± 2°, Figure S3) were prepared by immersing the pre-cleaned quartz wafers in CAB-O-SIL TS720 in acetone suspension (10 g L^{-1}) for 4 min and then dried at 80 ℃ for at least 5 h.

Section S2. Stimulated Raman Spectroscopy measurements.

The Fe(III)-Oxalate chelates chemical imaging was based on a conventional SRS microscope that our previous work has detailed.[1,2] Two synchronized pulse train of 80 MHz was produced by a commercial femtosecond laser system (Insight DS+, Spectra-Physics, USA). The Stokes beam served the fixed fundamental output of 1040 nm and the pump beam employed the tunable optical parametric oscillator (OPO) output of 680-1300 nm. The pulse durations of the Stokes beam and pump were stretched and chirped to several picoseconds by passing through SF57 glass rods in order to obtain high spectral resolution (~13 cm^{-1}). The intensity of the 1040 nm Stokes beam was adjusted by an electro-optical modulator (EOM, EO-AM-R-20-C2, Thorlabs) to 1/4 of the laser pulse repetition rate. The two laser beams overlapped temporally and spatially through a dichromic mirror (DM) and then transmitted into a laser scanning microscope (FV1200, Olympus) equipped with galvo mirrors for raster scanning. The combined beams were focused onto the microdroplet by a 60× water immersion objective lens (Olympus, UPLSAPO 60XWIR, NA 1.2). The simulated Raman loss (SRL) signal was optically filtered (CARS ET890/220, Chroma), detected by a homemade back-biased photodiode (PD) and demodulated with a lock-in amplifier (LIA) (HF2LI, Zurich Instruments) to feed the analog input of the microscope to form images with the size of 512 × 512 pixels. To ensure stable chemical imaging, the microdroplet was kept in a closed homemade chamber. The laser powers of the pump and Stokes pulses that samples used were 20 mW and 30 mW, respectively. In order to match the pixel dwell time, the lock-in time constant was set at 2 μs. All images were taken in transmission mode.

Supplementary Figures

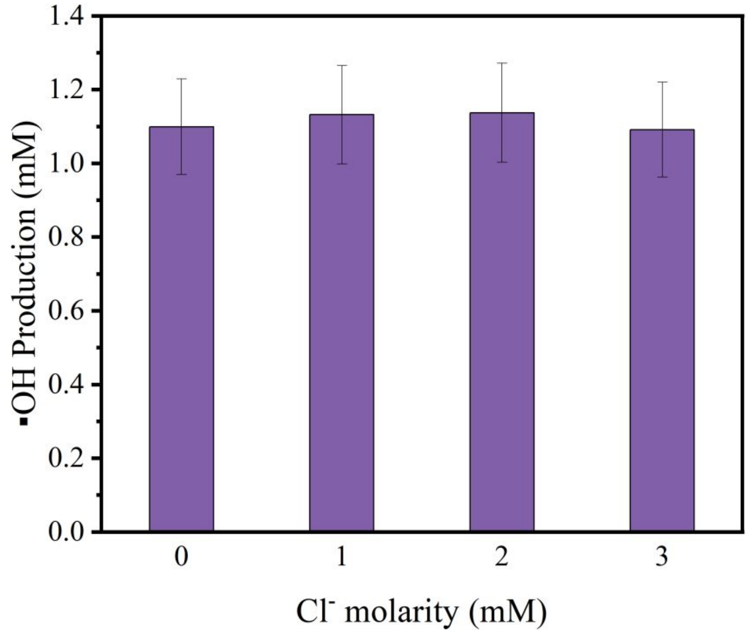

Figure S1. The influences of Cl^- on hydroxyl radicals generation by Fe(III)-Oxalate complexes photolysis in aerosol microdroplets. Experimental conditions: pH 3.5; Reaction time, 30 minutes.

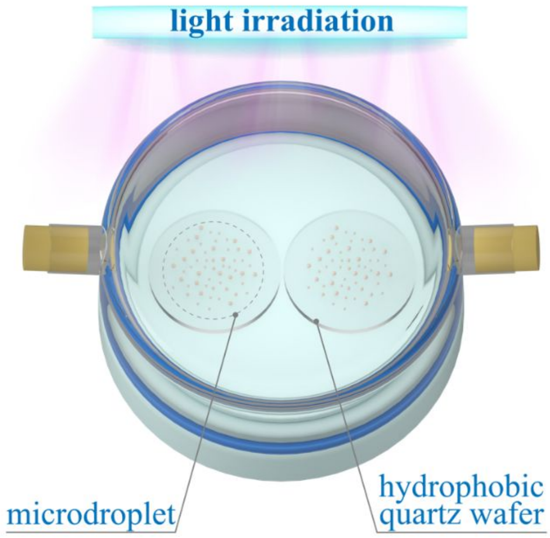

Figure S2. Schematic illustration of the custom-designed environmental chamber for microdroplet photoreaction.

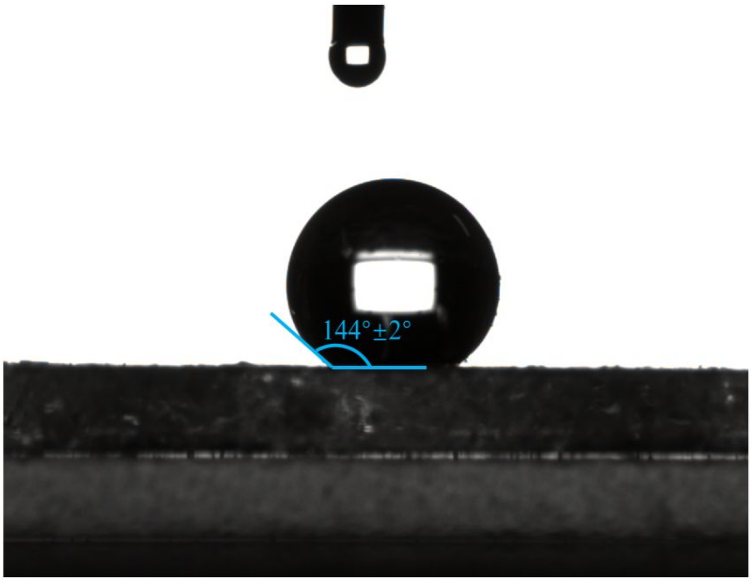

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Figure S3. The water contact angle of super-hydrophobic substrate.

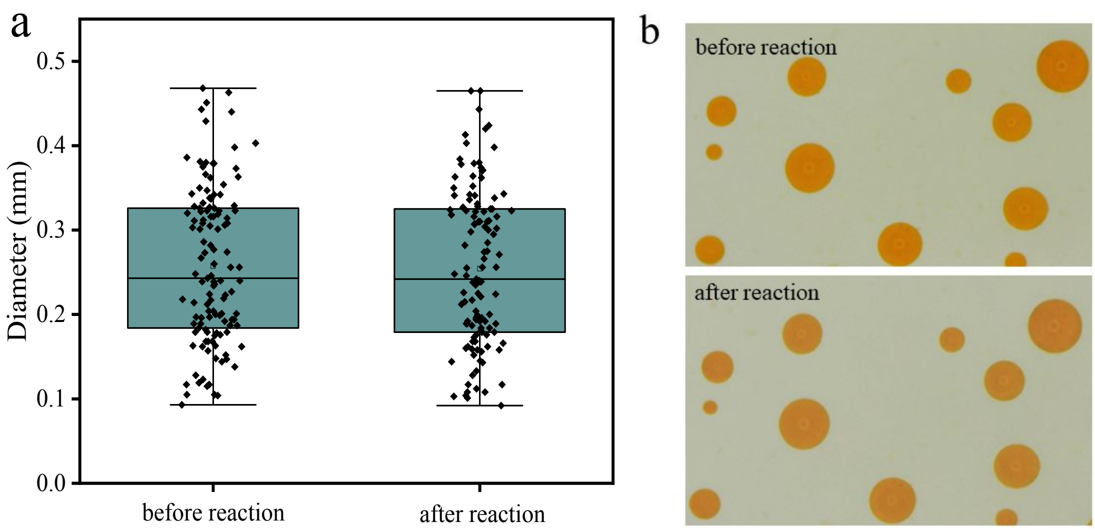

Figure S4. (a) Box plot (mean, median, interquartile range, and outliers) of microdroplets diameter before and after reaction. (b) A photo of the microdroplets before and after reaction.

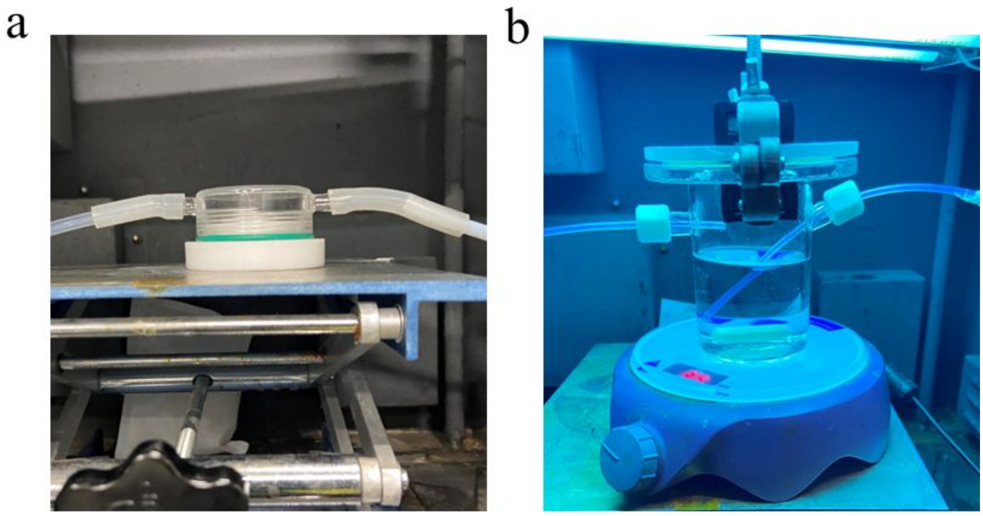

Figure S5. The experimental setup for sulfate formation in (a) microdroplets and (b) bulk solution.

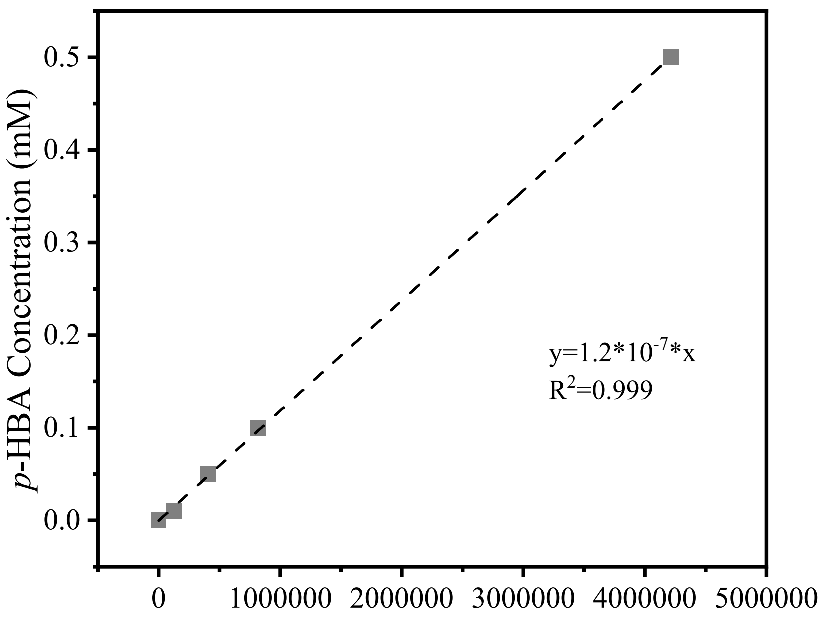

Figure S6. Fitted standard curve of *p*-HBA concentration.

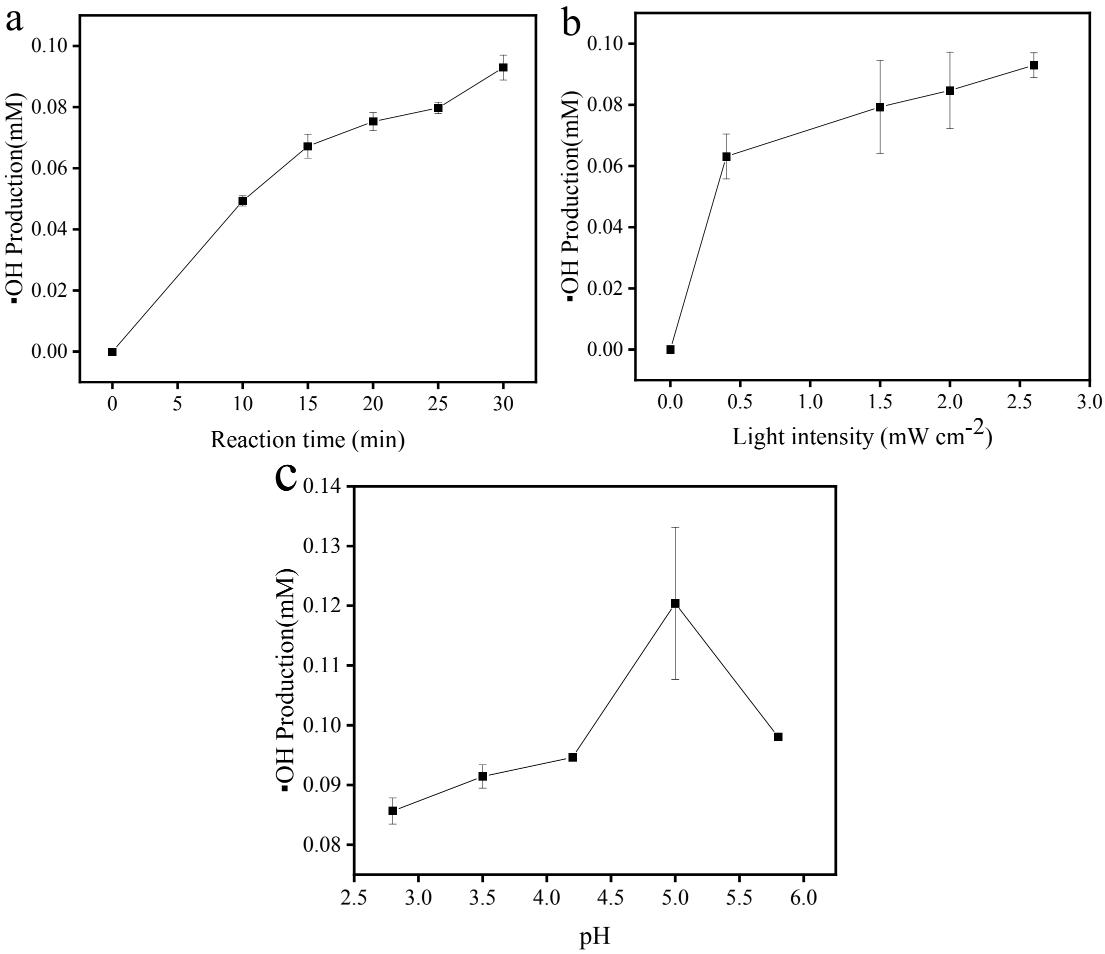

Figure S7. Hydroxyl radicals generation by Fe(III)-Oxalate complexes photolysis in bulk phase. (a) Influence of light irradiation time; (b) Impact of light irradiation intensity; (c) Effect of solution pH. Experimental conditions: [Fe(III)] = 500 μM; [Oxalate] = 3.5 mM; pH 3.5; Reaction time, 30 minutes.

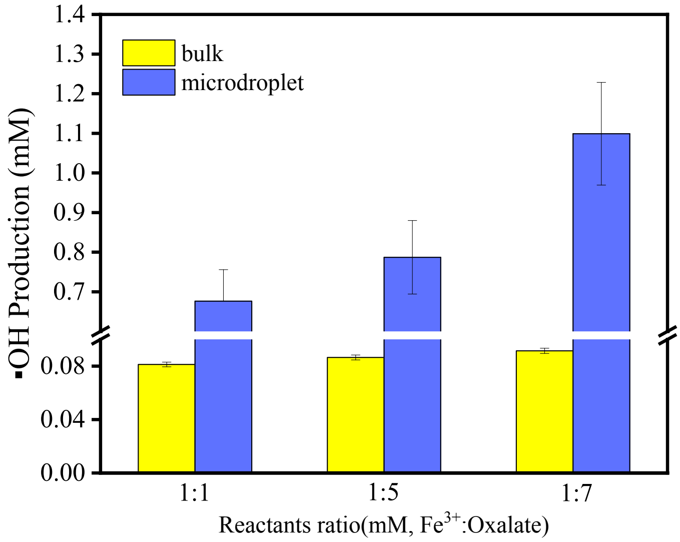

Figure S8. Influence of oxalate/Fe(III) molar ratio on hydroxyl radicals generation by Fe(III)-Oxalate complexes photolysis in aerosol microdroplets and bulk phase. Experimental conditions: pH 3.5; Reaction time, 30 minutes.

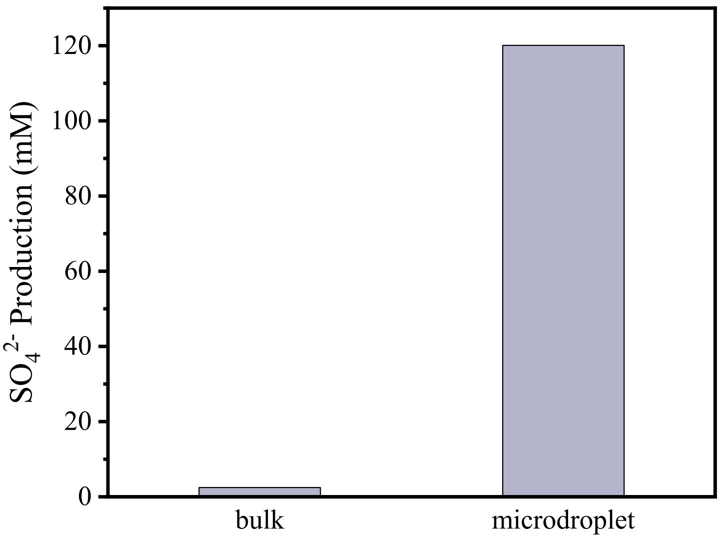

Figure S9. Sulfate formation in microdroplets and bulk solution under light irradiation with low concentration SO_2. Experimental conditions: [Fe(III)] = 500 μM; [Oxalate] = 3.5 mM; pH 3.5; [SO_2 concentration] = 1 ppm; Reaction time, 20 minutes.

References

1 Ao, J. P.; Feng, Y. Q.; Wu, S. M.; Wang, T.; Ling, J. W.; Zhang, L. W.; Ji, M. B. Rapid, 3D Chemical Profiling of Individual Atmospheric Aerosols with Stimulated Raman Scattering Microscopy. *Small Methods* **2020,** *4*, 1900600.

2 Ao, J. P.; Fang, X. F.; Miao, X. C.; Ling, J. W.; Kang, H.; Park, S.; Wu, C. F.; Ji, M. B. Switchable stimulated Raman scattering microscopy with photochromic vibrational probes. *Nature Communications* **2021,** *12*, 3089.